

- 6 (a) (i) diffraction bending/spreading of light at edge/slit
this occurs at each slit B1 B1 [2]
- (ii) constant phase difference between each of the waves B1 [1]
- (iii) (when the waves meet) the resultant displacement is the sum of the displacements of each wave B1 [1]
- (b) $d \sin \theta = n \lambda$
 $n = d / \lambda = 1 / 450 \times 10^3 \times 630 \times 10^{-9}$ C1
 $n = 3.52$ M1
hence number of orders = 3 A1 [3]
- (c) λ blue is less than λ red M1
more orders seen A1
each order is at a smaller angle than for the equivalent red A1 [3]
- 7 (a) thin paper reduces count rate hence α B1